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Learning Project Presentation on OCR for handwritten receipts

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INTRODUCTION

From scanned content to structured information.

As businesses and individuals increasingly rely on digital records, receipts and documents are often stored as images rather than usable data. This creates a need for systems that can interpret visual content and transform it into information that can be digitally processed.

- 🎯 **Visual documents** → Difficult for computers to process directly
- 🎯 **OCR + Computer Vision** → Enables automated text extraction
- 🎯 **Structured data** → Makes information easier to process and analyze

PROBLEM DEFINITION

Physical receipts are difficult to store, search, and analyze. Handwritten text can lead to inconsistent OCR results, while important information is often hidden in unstructured text. Manual extraction is also time-consuming and can increase the risk of human errors.

🚫 **The challenge:** Converting visual information into usable digital text

💪 **The Issue:** Manual processing is time-consuming and error-prone

| **Storage**

Physical receipts are easily misplaced or damaged.

| **Extraction**

Unstructured and handwritten content is difficult to process accurately.

| **Manual Effort**

Data entry is slow and susceptible to human error.

OBJECTIVE

- Extract text from images and scanned documents.
- Reducing the effort involved in manual transcription.
- Making visual information easier to process, search, and reuse.

SCOPE

- Digitization of physical and scanned receipts through OCR
- Preprocessing of receipt images for improved recognition
- Extraction and structuring of key receipt information

WHY ?

- Automates receipt data extraction, reducing manual effort
- Improves OCR accuracy through image preprocessing
- Converts raw text into structured, usable information
- Provides a simple and accessible web-based solution

WORKFLOW

- Input: Receipt images or scanned documents
- Processing: Image preprocessing and OCR-based text recognition
- Extraction: Identification of relevant receipt information
- Output: Structured, searchable, machine-readable data
- Interface: Simple web-based upload and result display



TECH STACK

Core Processing

- Python
- Tesseract OCR
- OpenCV
- Pytesseract
- NumPy
- JSON

Backend

FastAPI + Python

- REST API
- OCR pipeline integration
- Request & response handling

Frontend

React + Vite

- Document upload
- Results visualization
- Responsive user interface

CONCLUSION

The proposed OCR-based system provides an efficient approach to digitizing receipts by combining image preprocessing, OCR, and information extraction. It reduces the dependence on manual data entry while transforming unstructured receipt content into structured, searchable, and machine-readable information. With its web-based interface and modular technology stack, the system offers a practical and scalable solution for streamlined receipt processing.

Thank You